```python
def sum_n(n):
    if n == 1:
        return 1
    return n + sum_n(n - 1)

n = 10

print("SUM OF NATURAL NUMBERS")
print("---------------------------")
print("N =", n)

for i in range(1, n + 1):
    print(i, end=" ")

print("\n")

result = sum_n(n)

print("Sum of numbers from 1 to", n, "=", result)
print("Formula: N * (N + 1) / 2")
print("Verification:", n * (n + 1) // 2)
```

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/user/Downloads/AIES.py ==========
========
SUM OF NATURAL NUMBERS
---------------------------
N = 10
1 2 3 4 5 6 7 8 9 10

Sum of numbers from 1 to 10 = 55
Formula: N * (N + 1) / 2
Verification: 55
>>>
```